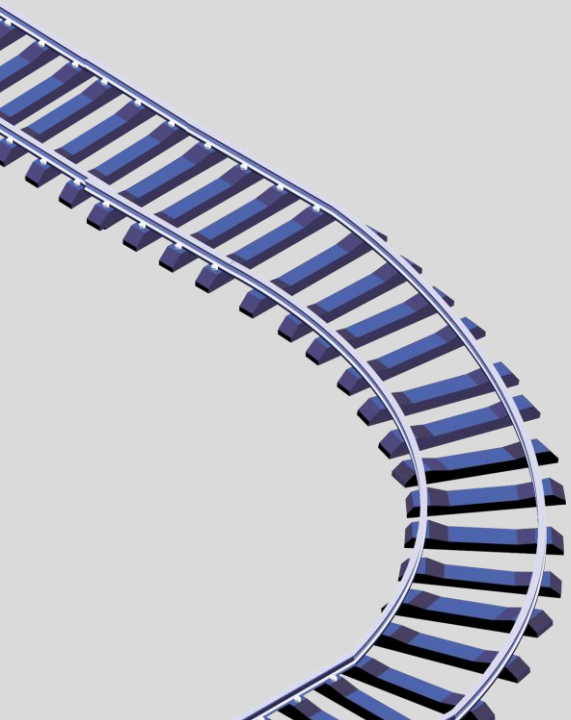
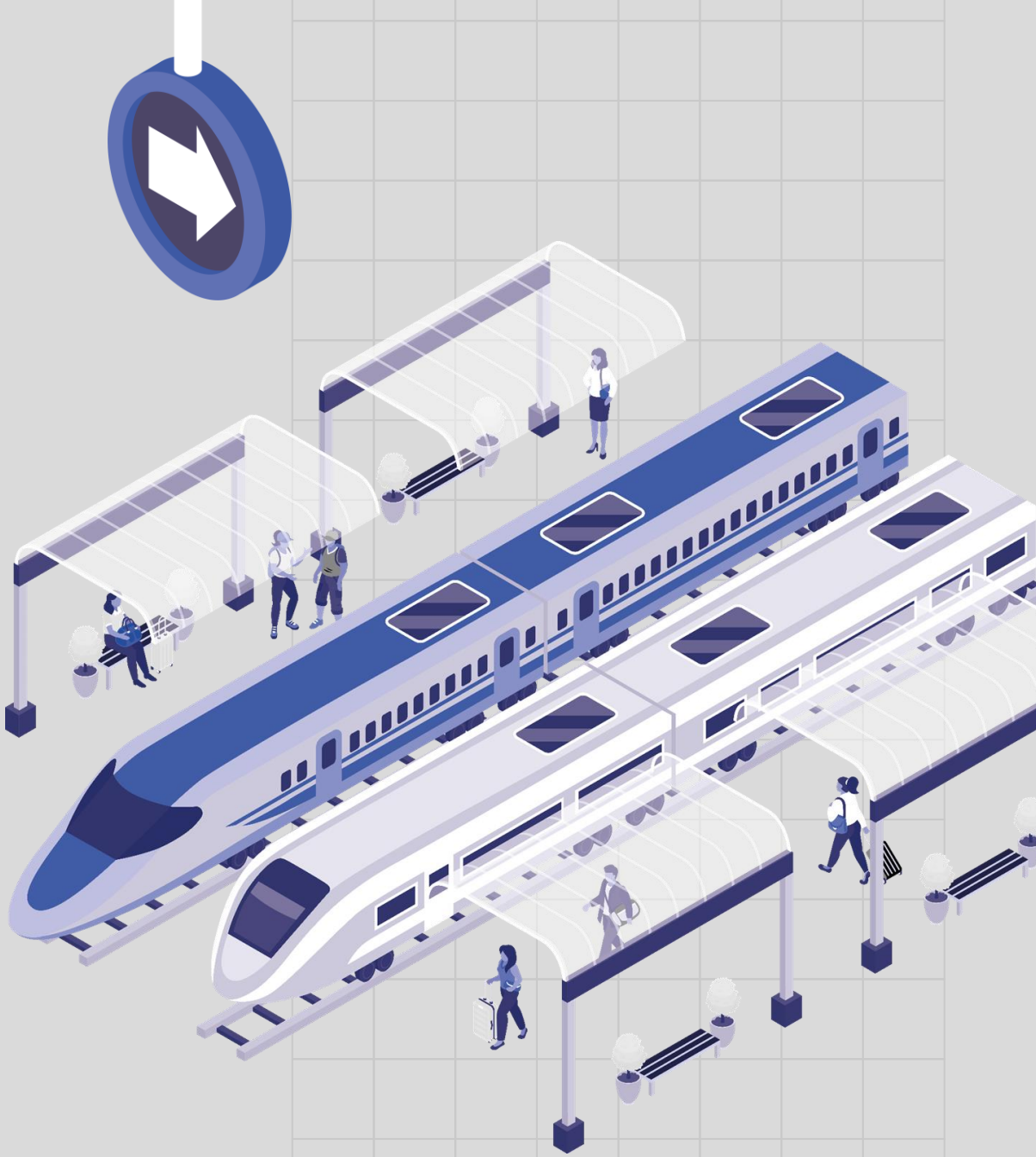
Rail Persona Analysis

By: Joshua Giese

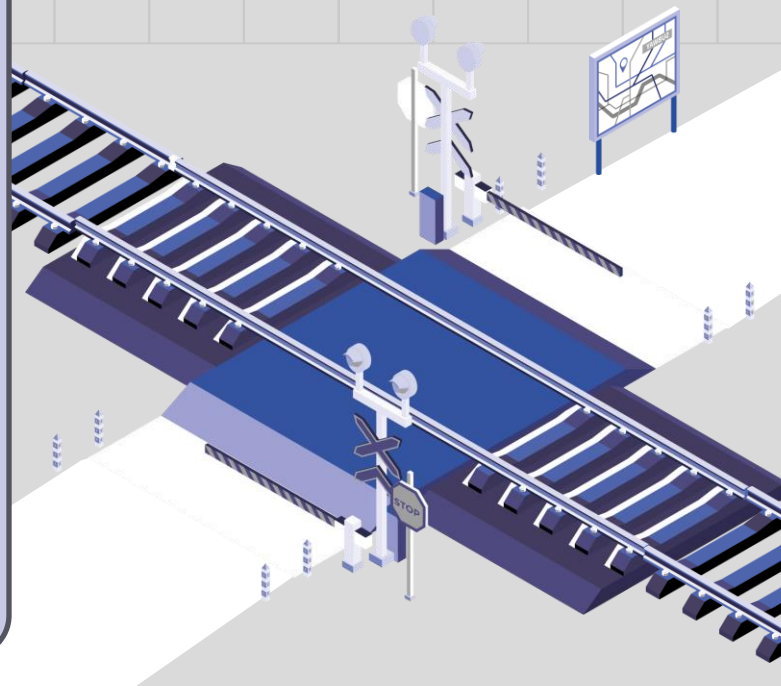




Administration & White-Collar Work:

Roles/ Jobs: Management, Legal, Regulatory, IT, Corporate Communications, Finance, Government Relations, Etc.

- Approximately 85% of Class I railroad employees and around 60% of non-Class I (smaller) railroad employees are represented by a labor union
- Most train companies are shifting towards “Precision Scheduled Railroading (PSR)” leading to drastic cuts in labor forces and significant increases in train length. Instead of acquiring more assets, this model focuses on efficiency and utilization of current assets. Train companies utilizing this often have trains upwards of 10,000 to 15,000 feet (2 to 3 miles long) though the average is around 6,000 feet.



Name: Arney Applehoods

Location: Canadian National Railway

Age: 46

Industry: Administration

Title: Director of Operations

Arney starts his day by opening his email, reviewing regulatory updates. He must then communicate with another department to start a lobbying campaign to gain more grants for infrastructure upgrades. He then must coordinate with signaling to start the rollout of a new software that will help manage the railroads better

Interest:

- Ways to improve efficiency within compliance requirements
- Modernizing digital systems such as in IT and Business
- Securing grants for infrastructure and upgrades

Challenges:

- Interpreting complex regulatory changes in the short term
- Keeping up with technological advancements
- Operating under tight budget restrictions
- Effectively communicating between all departments and stakeholders

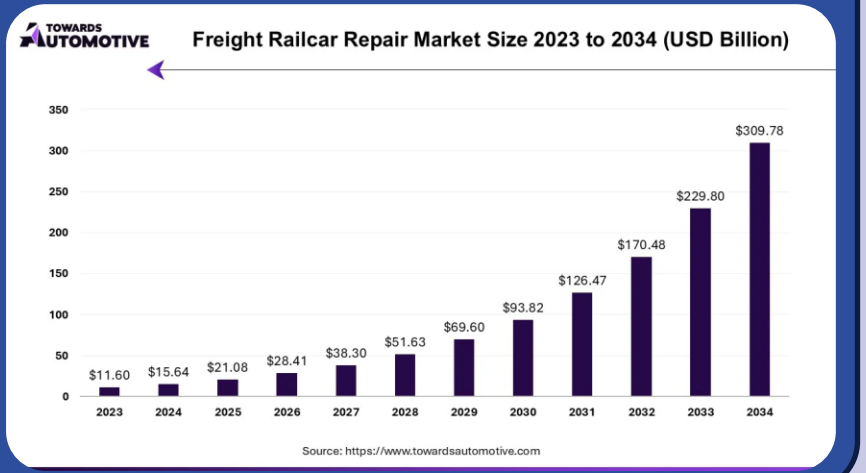
Desired Outcomes:

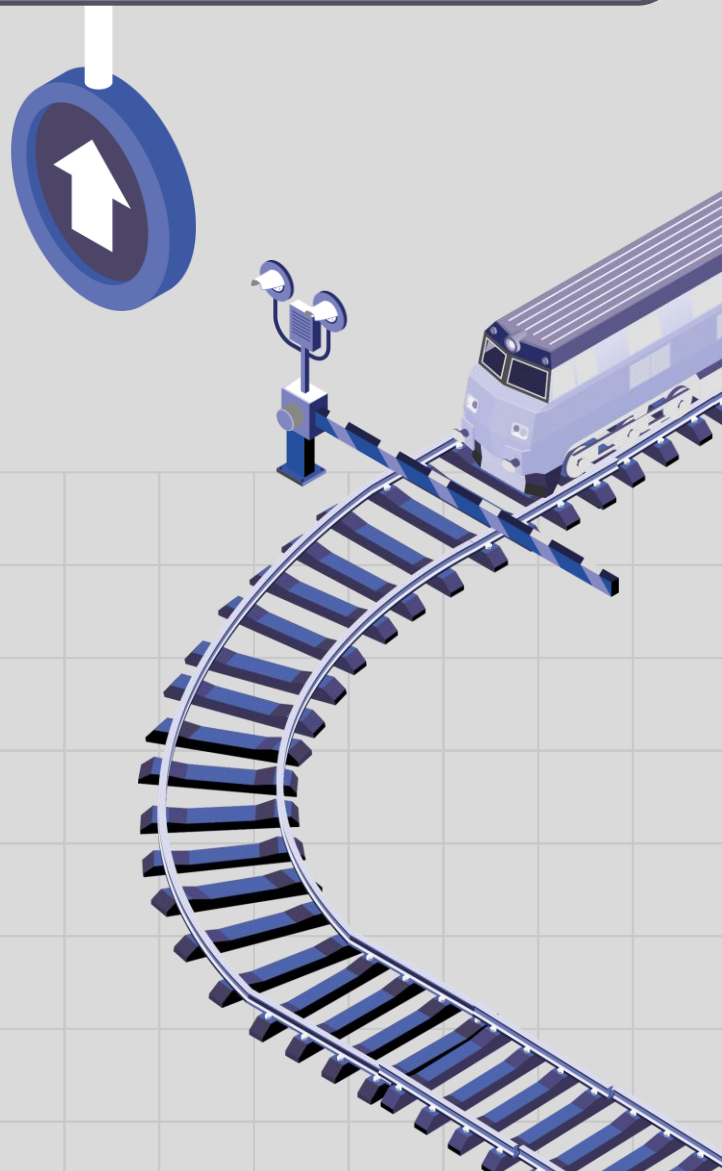
- Management of Regulatory Changes and compliance
- Digital Maintenance
- Cost control and efficiency
- Government grants and infrastructure spending

Mechanical: (Rolling Stock & Locomotives)

Roles: Car & locomotive mechanics, rebuilders, running-gear specialists, shop supervisors, Maintenance Engineers, Etc.

- FRA maintenance and inspection standards (Compliance manuals: https://railroads.fra.dot.gov/taxonomy/term/14801?page=0)
- Market trends: https://www.gminsights.com/industry-analysis/freight-railcar-repair-market
  - Increasing technical use, i.e., IoT sensors, AI, predictive maintenance software.
  - Increase in demand for mechanics and freight transport overall
- Projected market increases:
  - This is global prospects, however, the American markets are seeing some of the same trends.
  - More outsourcing compared to in-house work.





Name: Matthew Mechanichay

Location: Norfolk Southern

Age: 39

Industry: Mechanical

Title: Locomotive Mechanic Supervisor

Matt is the supervisor of a 10-person mechanical team. Due to recent tariffs and backups in shipping, a new break part that was on backorder is taking longer than expected to arrive at the shop. Today, he is checking diagnostics on a new program which has flagged a couple issues that they must fix before an engine can be operational again.

Interest:

- Increasing equipment uptime and reliability
- Access to parts and suppliers
- Diagnosing mechanical failures
- Meeting compliance and inspection requirements

Challenges:

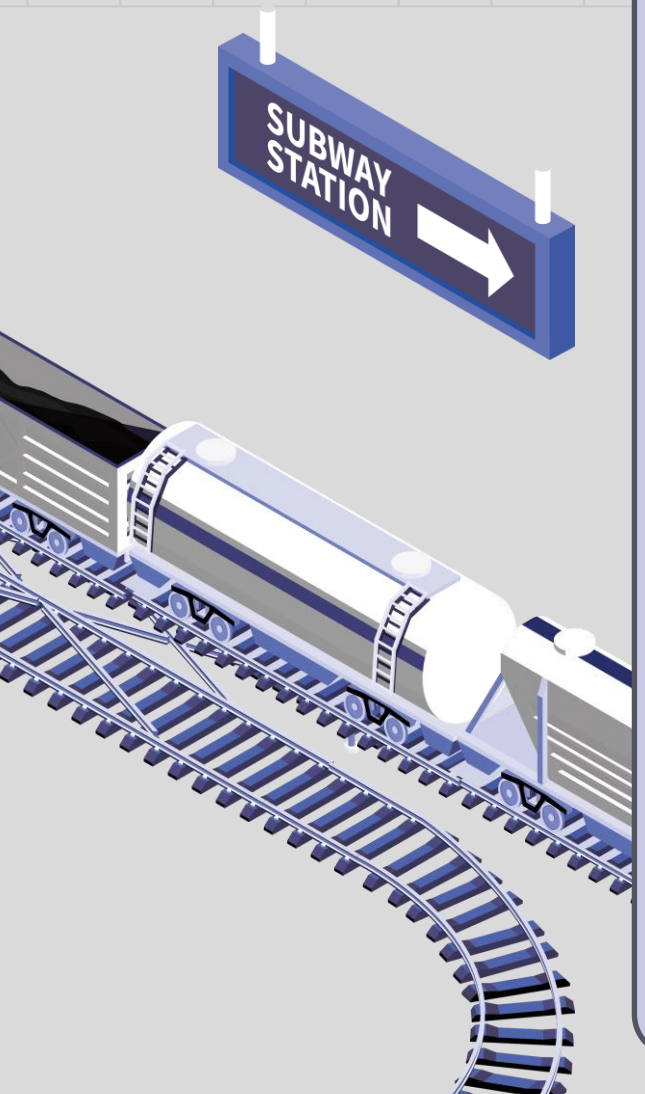
- Aging locomotives and delays/access to some parts
- Pressure to reduce turnaround time and reducing maintenance times
- Maintaining reliability and safety

Desired Outcomes:

- Locomotive upgrades and maintenance
- Smooth inspection process, compliance, and safety
- Predictable schedules for maintenance, repair, and upgrades
- Everything must be alright alright alright

Engineering: (Track, Bridge, Structures)

Roles: Track engineers, bridge & building engineers, roadway workers, civil engineers, surveyors, structures and MOW (maintenance-of-way) teams.



- An increasing trend towards use of AI and drones for assessing track and infrastructure repair. This is especially significant due to the U.S. train network comprising of over 100,000 bridges or about one bridge per 1.4 miles of track

Name: Edward Estabon

Location: BNSF Rail

Age: 28

Industry: Engineering

Title: Civil Engineer

Edward oversees the almost 400 miles of track and 18 bridges in Washington and Oregon that are managed by BNSF. After weeks of inspections, he finds that there is erosion near one of the bridges threatening damage or worse if not tackled within the next few years. They temporarily reinforce this section while he waits for approval and funding from the higher ups.

Interest:

- Preventative asset maintenance, design, and inspections for long-term
- Access to long-term project funding and explaining its importance to management
- Maintaining industry engineering standards

Challenges:

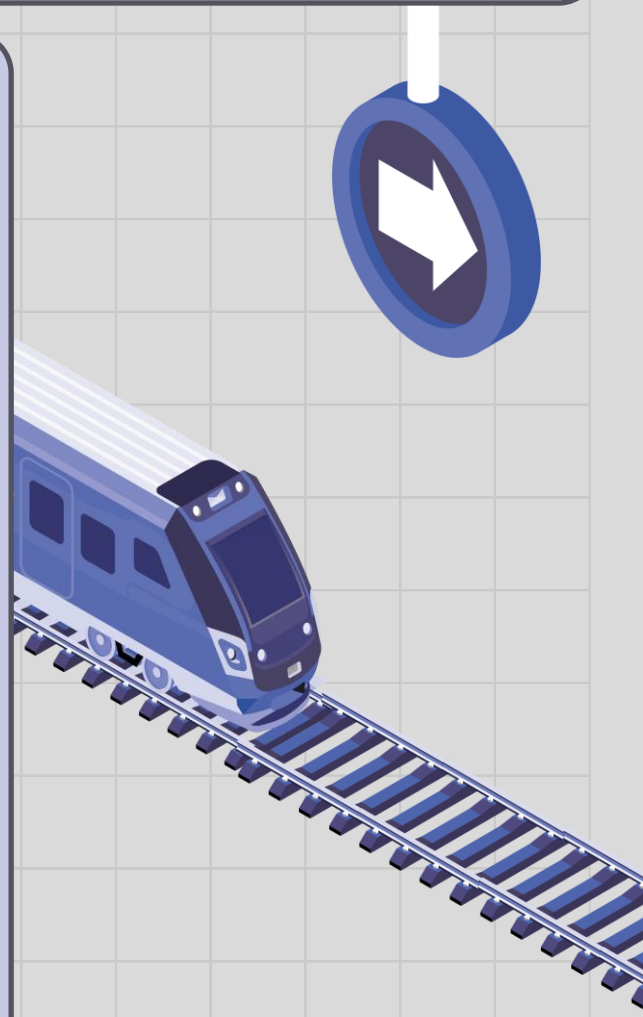
- Juggling Maintenance, Upgrades, and new infrastructure plans
- Working in remote environments for track maintenance
- Coordinating work in windows without train traffic to prevent downtime

Desired Outcomes:

- Safe and up to date rails and buildings
- Predictable funding for projects
- Reduce the need for emergency repairs

Logistics & Network Operations:

Roles: Train dispatchers, planners, network optimization analysts, supply chain coordinators, procurement, environmental compliance.



Name: Larry Two-Shoes Diddly the Third

Location: Union Pacific

Age: 58

Industry: Logistics & Network Operations

Title: Network Planner

A Tornado has touched down in Missouri wiping out miles of track and halting freight that was on its way to Kansas city. Larry notifies emergency engineering crews and reroutes trains to another path which will take longer than before. He then continues to help shift and redistribute traffic around the area.

Interest:

- Real time train tracking for own assets and other companies on the same lines
- Efficient yard and transportation utilization
- Transportation strategies to reduce travel time
- Rail regulation compliance

Challenges:

- Unpredictable delays and congestion
- Coordinating between companies and departments
- Pressure to reduce travel times and increase load per freight car
- Making sure trains don’t run out of fuel between stops

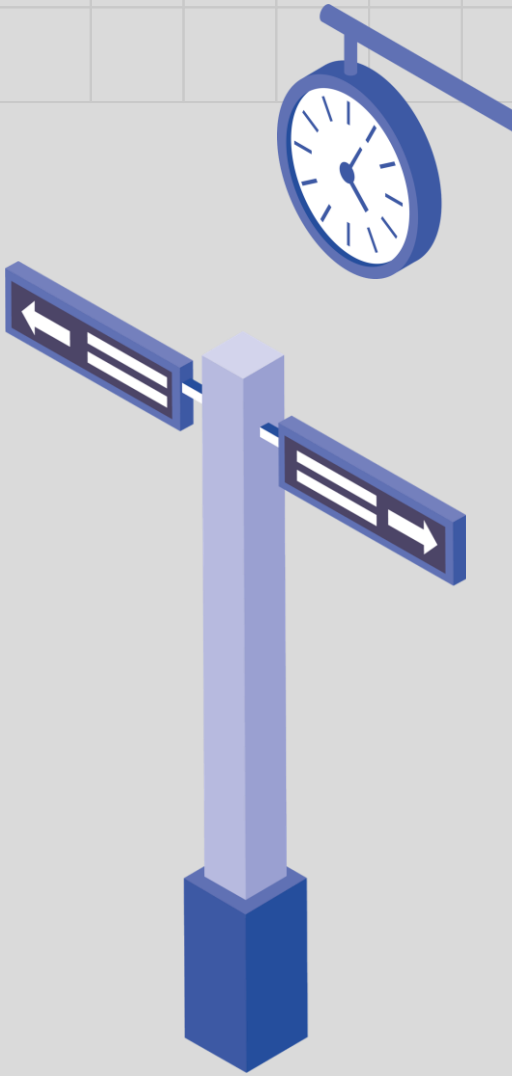
Desired Outcomes:

- Optimized network with little disruptions
- Organized/optimized scheduling of cargo
- Service reliability and efficiency

Signaling & Communications:

Roles: Signal maintainers, dispatch system engineers, PTC/CBTC specialists, communications engineers, Security and safety

- Positive Train Control (PTC) systems are designed to prevent train-to-train collisions, over-speed derailments, incursions into established work zones, and movements of trains through switches left in the wrong position.
  - Overview and Policy Issues: https://www.everycrsreport.com/reports/R42637.html



Name: Sammy Smithington

Location: CSX Transportation

Age: 35

Industry: Signaling & Communications

Title: Chief of Signal Maintenance

Sammy is currently on site to diagnose some problems with a newly installed PTC device. He goes into the conductor's cabin, runs the fix, and returns to helping install some new train signals at a freshly completed crossing.

Interest:
- Secure and modern communications network
- Real time diagnostics tools and control systems
- Regulatory compliance (FRA/PTC)

Challenges:
- Aging Infrastructure and changes in remote areas
- Protecting assets from technical threats and hackers
- Preventing system failure
- Coordinating signal upgrades across rail lines

Desired Outcomes:
- Regulatory and safety compliance
- Scalable and futureproof communications and signaling systems
- Little to no disruptions of service due to signal failure